Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	404 Brain Not Found
Project / Solution Name	TrendPilot AI
Name (Leader)	Teng Ray Tzong
Email & Phone (Leader)	+60 11-31317873
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	Not publicly deployed at the time of submission
Code Repository URL (public / access granted to judges)	https://github.com/runforfun19/trendpilot-ai.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

TrendPilot AI uses a frontend-backend architecture. The frontend is built with Next.js, React and TypeScript. Users enter skills or career interests through the web interface, which sends an HTTP POST request containing JSON to the Python FastAPI backend.

The backend validates the input using Pydantic, constructs a structured prompt, and calls the Groq API through the OpenAI-compatible Python client. The configured model defaults to openai/gpt-oss-20b, although the runtime model depends on the GROQ_MODEL environment variable. The backend parses the model response, normalises career-role fields, bounds match scores to 0–100, and returns structured JSON to the frontend.

The frontend displays career suggestions, match estimates, matched skills, skill gaps, learning pathways and SDG connections. A separate GET /market-trends endpoint returns a static snapshot of Malaysian employment statistics labelled with DOSM as the source.

No database, authentication service, or public cloud deployment was identified in the submitted source files. The current repository includes local development instructions. API credentials are expected to be supplied through the backend environment file rather than embedded in source code.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python — runtime version to confirm	Backend logic and API endpoints	Python Software Foundation License
FastAPI — installed version to confirm	REST API framework	MIT
Uvicorn — installed version to confirm	ASGI server	BSD-3-Clause
Pydantic — installed version to confirm	Input validation and data modelling	MIT

Name & Version	Purpose in Project	License
OpenAI Python client — installed version to confirm	OpenAI-compatible client used to call Groq	Apache-2.0
python-dotenv — installed version to confirm	Load environment variables	BSD-3-Clause
Next.js 16.4.2	Web framework	MIT
React / React DOM 19.3.0	Frontend UI	MIT
TypeScript 5.x	Frontend type checking	Apache-2.0
Tailwind CSS 4.x	UI styling	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Groq API — OpenAI-compatible chat completions endpoint	Generate career-role suggestions, estimated match scores, skill gaps, learning pathways and SDG links	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Local development environment — VS Code / local machine	Develop and run the frontend and backend	Local; no cloud plan
Groq API	External LLM inference	No cloud plan
Database service	None identified in current source code	N/A
Public cloud hosting	Not yet verified in the uploaded project files	N/A

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
openai/gpt-oss-20b	Groq API hosting the model	Pre-trained model used for inference; no project-specific fine-tuning identified	Apache-2.0 for the model; Groq API terms apply

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Employment Statistics, First Quarter 2026	DOSM; https://www.dosm.gov.my/portal-main/release-content/employment-statistics-q12026	Public	Verify applicable source terms	N
User-entered skills / career interests	Input submitted through the application	User-provided input; not a training dataset	Not applicable as a project-owned dataset	y

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
ChatGPT — exact model/version to confirm	Code assistance, debugging guidance, technical documentation and/or pitch preparation, according to actual use	Moderate or Major, based on the actual contribution

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team reviewed AI-assisted code and project materials against the submitted source code and the intended functionality. API integration, input validation, response parsing, and error handling were checked during local testing. AI-generated career suggestions and match scores are treated as estimates rather than verified facts. The team remains responsible for checking outputs, verifying external data sources, identifying limitations, and correcting issues before submission. Features that are proposed but not implemented are explicitly identified as future improvements.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☑ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application accepts user-entered skills and career interests and sends them to the backend for analysis. The backend forwards the input to the Groq API to generate career recommendations. Users should avoid entering names, contact details, CVs, or other sensitive personal information.

The submitted source code does not show a database or a persistent user-profile storage mechanism. However, the application does not establish that data is never processed or retained by the external API provider. Any retention by the provider is subject to its applicable terms and policies.

The Groq API key is expected to be stored in a local backend environment file, which is excluded from version control through the project's ignore rules. Before public deployment, the team should verify that no credentials are exposed, review the provider's data-handling terms, and document any applicable retention period. The current source does not establish a complete retention policy for third-party processing.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The main limitations are LLM-generated inaccuracies, potentially biased career suggestions, unstable or unavailable API access, and limited personalisation when users provide insufficient skills information. Match scores are model-generated estimates and have not been validated as employment probabilities.

The employment indicators are hard-coded published statistics rather than a live, automatically updating feed. They describe national employment conditions and do not directly establish demand for particular technologies or skills. The current prototype also does not implement explicit alternative future scenarios or a validated forecasting model.

Mitigations include structured JSON output, backend input validation, score bounds, response normalisation, and explicit notes about the limitations of match scores and employment data. Further improvements should include verified and dated external data sources, systematic testing with representative profiles, bias and accuracy reviews, clear error handling, and transparent source attribution.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] 

[Date]24/10/2026